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Computer Architecture : Computer system by specifying its Parts and their Relations

- (*) direct Impact on logical execution
- (*) What Computer Does?
- (*) Indicates Computer hardware
- (*) Architecture is Fixed First

Computer organization : operational units and their Interconnection that realize the Arch specification

- (*) You can change organization without changing Architecture

All Intel X86 Family → Same basic Architecture
→ but Differs organization

- (*) How it does it?
- (*) Reveals Computer Performance
- (*) Is Decided After the Architecture is Fixed

UNIVAC and ENIAC

- ① 20 Accumulators For + and -
- ② I/O From Punch Cards
- ③ multiplier, Divider, square root unit

Von Neumann Architecture

Central Arithmetical unit (CA) → arithmetic operations

Central Control unit (CU) → sequencing of operations

Memory (M) → Store numerical Data And Instructions

I/O devices → Interface between computer and user

IAS

Accumulator Register (AC) تخزين نتائج مؤقتة

Arithmetic Register (AR)

memory [RAM] 40 bit $\rightarrow$ 1024 word

Instruction 20 bits while one data value are 40 bit

(180) memory Buffer Register (MBR) $\rightarrow$

contain word to be stored in memory

hold word to be sent to or receive from memory or I/O

(180) memory Address Register (MAR)

Specific Address in memory of word for R or W

(180) Instruction Register (IR) :

Contains the 8-bit OP-code

(180) Instruction Buffer Register (IBR)

right hand Instruction

(180) Program Counter (PC) $\rightarrow$ contain Address of
next instruction to be fetch from memory

or

Keep track of Program Instruction stored in memory

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280
Accumulator (AC)280
Multiplier Quotient (MQ)

Temporarily operands

Temporarily operands

Most-Significant DigitsLeast Significant bit

OUT PUT of the ALU

results of ALU

The divisor for divisionQuotient from division280
PC → MAR → Memory → MBR → IR

Fetch Cycle → op-code of Next Instruction is loaded into IR

taken from IBR or from memory by loading word into the MBR and then down to the IBR, IR and MAR

Execute Cycle → Performed when the op-code is in IR

Neumann 3 part move ② ALU ③ control
un-conditional

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System Bus: Shared Pathway System made up of Collection of wires are grouped by function That mainly cares about the Communication the other component

only one device can send data
other devices simultaneously listen
only one device receives

Devices ① Master _{Send} ② Slave _{receives}

Synchronous Bus

clock (CLK) Send Sequence 1's and 0's

Clock signal is derived from the Master System Clock

Master Output valid → on the rising edge of CLK
stay valid → through the falling edge

Slave output (for read) valid by falling edge

Asynchronous Bus

NO Clock

Transmitters and receivers in all timing based on Control signal edges

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(40)

Hand Shaking Protocol

STROBE $\rightarrow$ Master to Slave
Slave read on falling edge

ACK $\rightarrow$ Slave to Master
Master reads on falling edge

Synchronous

Asynchronous

(Adv)

lower overhead
greater throughput

cheap $\rightarrow$ less hardware
doesn't require synchronization
Setup is faster (40)
 $\rightarrow$ Irregular Intervals

(dis)

Complex large overhead
expensive more Hw

Semi-Synchronous Bus

Everything is synchronized to bus clock, but the transaction take variable number of cycles

Wait States: Extra clock cycles between start and end due to late ACK

Slave Assert ACK to data is valid at clock edge

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Data Bus → EXchanging Data between, CPU, M and Peripherals
→ Bi-directional
→ Bus width → Number of wires

Address bus → Where the Information is being Send
→ only Direction
→ width Address = 2 lines

Control bus → Carries the Signal from the CU relating to Control and Co-ordination of Activities

Dedicated → bus used only for one Purpose

يستخدم لغرض واحد فقط

Multiplexed → Same line at different time for multiple Purposes

Fewer lines and chips → limited in Num Pins

Data and address appear on the bus at different times

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PCI Bus

① Data and address lines are multiplexed

②

PCI Bus Peripheral connected on this bus to control and transfer data directly

DMA → Direct Access Memory

PCI

① higher throughput

lower I/O pin count

smaller physical footprint

better performance

Express bus is not pluggable

PCI Express Bus

1x Connector	36 Pin	High bandwidth I/O
4x	64 Pins	Server
8x	98 "	"
16x	164 "	Graphics Port

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universal Serial Bus (USB)

USB 1.0 original 127 Devices 12 Mbps

USB 1.1 Fullspeed 12 Mbps

USB 2.0 hi Speed 480 Mbps

Capable of supporting USB 1.0,
1.1

Interrupting

Interrupt: External Event to Currently Executing Process
That Change in Normal Flow

Interrupt → high Priority condition

Interrupts handler or Interrupt Service Routine /
CPU suspend its current activities, saving state
and executing function

External Interrupts → I/O devices

Internal Interrupts (Traps)

- ① some illegal operation
 - ② erroneous use of data
 - ③ Exception that has been caused by program itself
- ↳ CPU timer

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Software Interrupts

only during the execution of instruction

Switch from user mode to supervisor mode

Multiple Interrupts

Multiple Interrupt lines

cpu picks the " " with highest priority

Priorities are assigned during cpu design phase

SW Polling

cpu detects interrupt

branches to ISR to poll each determine caused interrupt

cpu raises TEST I/O and place address I/O order of polling

Daisy Chain

I/O modules share common interrupt request line
vector on data lines

vector address of I/O modules or
some other unique identification
vector as pointer

priority of module is determined by
position of the module in the daisy
chain

صل على النبي

لا تنسوا أبي وجدتي من دعائكم

لا تنسوني من صالح دعائكم

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Bus Arbitration: more than one module may need control of the bus

Centralized ~~Star~~ Scheme

Single HW Device $\Rightarrow$ bus controller or arbiter allocating time

device maybe separate modules or Part of CPU

Distributed Scheme

No central controller

Each module contains access Control logic

and modules act together Share Bus

Different device Identification techniques

to identify the devices and to set

the priorities of the devices

Self Independent